

• PT Examination

1. Cognitive function

- Memory, orientation, conceptual reasoning, judgement should be checked
- Done through Mini-Mental Status Examination (MMSE)

<u>Test For</u>	<u>Score</u>
• Orientation	
Time, Day, Date, Month, Year	5
Place, Floor, City, State, Country	5
• Registration	
Name 3 objects and ask to repeat	3
• Attention and calculation	
Serial Subtraction (100-7-7-7-7-7)	5
• Repeat	
Recall 3 objects named above	3
• Language	
Name two words	2
Follow 3 step command	3
Copy intersecting polygons	1
Repeat therapist words	1
Follow written command	1
Write a sentence	1

2. Sensory examination

→ following sensations are checked:

- Superficial
 - Pain
 - Touch
 - Temperature
- Deep
 - Pressure
 - Vibration
 - Joint Position sense
- Cortical
 - Tactile localization
 - Stereognosis (object recognition)
 - Two point discrimination

3. Motor function

- Tone
 - checked by slow passive motion
 - Severity of rigidity can be validated by the level of resistance
- ROM
 - ROM of UL, LL and spine should be etc checked
- Bradykinesia

- Tremor - location, Persistence and Severity should be recorded
- Postural control and Balance
- Done by Berg balance scale

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14 item list - each item has 5 points (0-4)

This includes the following:

- Sitting to standing
- Standing unsupported
- Sitting "
- Standing → sitting
- Transfers
- Standing - eyes closed
- " - feet together
- Reaching forward - outstretched arm
- Retrieving object from floor
- Turning to look behind
- " 360°
- Placing alternate foot on stool
- Standing - one foot in front
- " on one foot

• Gait - All the gait parameters should be examined

• Fatigue - checked - Fatigue severity scale

- Coordination - Finger - Nose - Finger Test
 - Heel shin test
 - Tandem walking

3. chest assessment

- Breathing pattern
- Inspiratory - Expiratory ratio
- chest wall mobility
- RL

Hoehn

- Hoehn and Yahr staging of Parkinson's disease
- Stage 1 - Sign and symptoms on only one side
 - Mild symptoms
 - Symp. inconvenient but not disabling
 - Tremor of one limb
- Stage 2 - B/L symptoms
 - Minimal disability
 - Balance Normal
- Stage 3 - B/L involvement
 - Mild postural imbalance
 - Patient leads independent life.
- Stage 4 - B/L involvement
 - Postural instability
 - Patient req. help
- Stage 5 - Severe
 - Patient restricted to bed / chair

PT Management

- Patients are guided by Physiotherapist to exercise at a high intensity for 1 hr 4 times a week - 4 week

Eg: Reach left arm across the body to opposite side, keep hand open, Palm up, right leg fully extended, toe pushing into the floor.

- Repeat on other leg.

Relaxation

- Gentle rolling
- Rotational movt of limbs and trunk
- Stretching
- Positioning - Hook lying, Side lying ^{rolling} ~~rotation~~, lower trunk rotation
- Diaphragmatic breathing $\bar{c}$ exercise
- Lifestyle modifications

Flexibility

- Active assistive
- Passive movement
- PNF $\bar{c}$ resistance
- Should be performed min 2-3 days / week

Ideally 5-7 days / week

- DNF Stretching (Contract Relax) - 8 s contraction followed by 10-30 s assistive stretch
- Yoga poses
- UE - D₂ flexion patterns



Promote upper trunk extension and counteract kyphosis

- unilat bridging ± trunk rotation
- BL "
- High kneeling ± ant pelvic translation

Stretch tight hip flexor and strength Spinal and hip ext

- UE - D₁ extension pattern - counteract Flexed and Adducted UE
- Positioning
- Resistive exercises
- MRP
- Sit to stand
- wt bearing ex.
- Ankle ex
- Mobilizing facial muscles
- Backward stepping can be used to strengthen hip and spinal extensors and promote upright posture
- Pt ± PD experiences high no. of falling



Pt should be taught how to get up after fall



Quadruped creeping should be taught so that the patient is able to move to nearby support.

• Balance training

- Emphasize practice of dynamic stability tasks

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- weight shifting
- Alternate UL w/ bearing
- Reaching
- Axial rotation of Head and Trunk
- Axial rotation $\bar{c}$ reaching

- Sitting on Physio ball

- Squatting

- kneeling

- Half kneeling

- standing

- changing arm position - Arms folded cross chest

- Reaching etc

- changing foot " - feet apart

feet together

- overhead arm clapping

- Heel raises

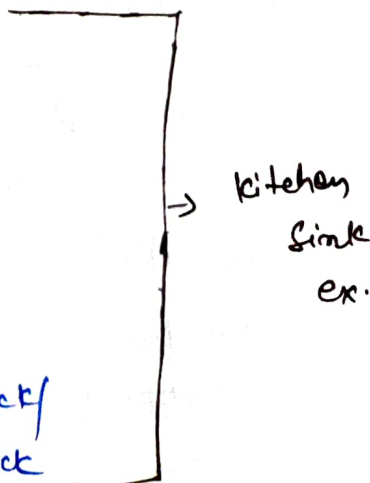
- Toe standing

- Toe walking

- Partial wall Squats

- - chair rises

- Single limb stance $\bar{c}$ side kick
Back kick



- - whole body vibration seated in a Physio acoustic chair

- Locomotor / Gait training

- Verbal command

- Visual and auditory cues

- Task specific training will be more beneficial

- Pulmonary rehab

- 4 main pulmonary complications in PD are.

- upper airway obstruction

- Restrictive disorder

- Aspiration Pneumonia

- Diaphragmatic breathing

- Postural drainage

- - Air shifting techniques

- vibration and shaking

- Home Plan (HEP)

- Make the patient understand about the importance of self exercises

- ~~Focus~~ Early morning warmups

- Stretching and Strengthening - Supine, sitting, standing

- Hanging by hand using overhead bar - Prevent banding & kyphosis

- Corner wall stretches